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Assessment of factors influencing PM mass concentration measured by gravimetric & beta attenuation techniques at a suburban site

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HIGHLIGHTS

- Discrepancies between BA and GM PM2.5 measurements are due to available water vapor.
- Correlation between BA and GM data depends on filter material.
- The variability of the ratio between BA and GM also depends on semi volatile species.
- It is suggested that methods other than heating are best for drying the air sample.

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ABSTRACT

Near real-time atmospheric particulate matter (PM) monitors are extensively used in air quality networks given their ability to provide continuous measurements with minimal attention by the operator. Their principle of operation is based on measurement of a physical parameter that is quantitatively linked to the PM mass concentration. Significant discrepancies between these measurements and those obtained by the reference gravimetric method, conducted in regions with diverse climatic conditions, have been reported in the literature. In this study we compare systematic $PM_{2.5}$ and PM_{10} gravimetric (GM) and beta attenuation (BA) measurements performed at a suburban site in Athens, Greece, over a period of 4 years (2009–2012). In general, BA and GM datasets exhibited similar temporal variation for both PM size fractions. An overestimation of the BA measurements, which was ~30% for the $PM_{2.5}$ and ~10% for the PM_{10} data, was observed. Good linear correlations between GM and BA data were observed, with estimated Pearson coefficients being 0.79 for the $PM_{2.5}$ and 0.85 for the PM_{10} measurements. The respective fitted equations through the entire dataset were $BA = 0.71\ GM + 6.2$, and $BA = 0.77\ G\ M + 4.1$. Better correlation between GM and BA measurements was observed during the cold rather than the warm period. Discrepancies between BA and GM $PM_{2.5}$ measurements increased with increasing available water vapor, suggesting that the aerosol bound water has a strong effect on the measurements. The effect of filter material used for GM measurements (i.e., quartz, glass fiber, or Teflon) was also examined for the $PM_{2.5}$ dataset. Best correlation between BA and GM data was observed when glass fiber, which is incidentally the material of the BA filter tape, was used in the GM measurements. When the BA to GM relationship was examined by further categorizing the data by the season (i.e., cold and warm period) for different filter types, the relationships that were fitted to the data for the two seasons were similar when Teflon filters were used, but quite diverse when quartz and glass fiber filters were employed in the GM measurements. Finally, the variability of the ratio between the two measurement techniques was found to be potentially dependent on the availability of the volatility or stability in the aerosol phase of species such as ammonium nitrate.

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1. Introduction

Inhaled particles can cause a number of respiratory, cardiovascular, circulatory and nervous diseases including chronic bronchitis, and asthma, alveolar inflammation, cardiac arrhythmia, as well as hypertrophy (Seaton et al., 1995; Schwarze et al., 2006). It is not surprising, therefore, that particulate matter (PM) pollution in urban environments has been associated with increased morbidity and mortality rates in several epidemiological investigations (e.g., Katsouyanni et al., 2001; Puett et al., 2011; Wang et al., 2013; Ostro et al., 2015). The documented detrimental effects of particulate matter pollution to human health have drawn attention to the need for integrated mitigation strategies, including the establishment of air quality guidelines, continuous and accurate monitoring of PM concentrations and effective control of population exposure.

A number of air quality directives have been set by the US Environmental Protection Agency (US-EPA) and the European Commission (EC) since the early 1980's. The first air quality guidelines established by the EC introduced threshold values for "black smoke" which was determined by measuring the light reflectance from PM loads collected on paper filters (Wolff and Perry, 2010). Since then, the EC legislation has been reviewed many times, following the evidence from epidemiological and toxicological studies. The most recent EC Directive (2008/50/EC) includes daily and annual limit values for ambient PM_{10} concentration, as well as an annual target value for $PM_{2.5}$.

Apart from setting the threshold PM values, another objective of the directives are to establish standardized techniques for measuring PM mass concentrations in order to ensure the quality and uniformity of the data. The use of near real-time PM monitors has been supported in US and European air quality networks over the past decades mainly due to their ability to provide continuous measurements with minimal human intervention. All types of PM gravimetric samplers, including the sampler recommended by EPA, can provide accurate measurements, considering a 10% variation of mean concentration (Yanosky and Maclntosh, 2001). In contrast to the gravimetric (GM) method, i.e. the reference method for measuring PM concentrations, there are no standardized techniques for near real-time observations. As a result, international policy-making bodies such as the EC Working Group on PM, support actions towards comparability and equivalence among methods for measuring PM mass concentrations (US-EPA, 1999; EC, 2010).

The most commonly used near real-time techniques for measuring PM are the Tapered-Element Oscillating Microbalance (TEOM) and Beta Attenuation (BA) method. Several studies have compared indirect/near real-time BA with direct GM particle mass concentration measurements and identified a number of parameters that lead to overestimation of the mass concentration determined by the former (e.g., Chang et al., 2001; Salminen and Karlsson, 2003; Lazaridis et al., 2008; Shin et al., 2011, 2012). These differences often depend on meteorological conditions and therefore exhibit seasonal variability (Gębicki and Szymańska, 2012). In addition, variations in the aerosol chemical composition can induce differences between BA and GM measurements.

Considering that both meteorological conditions and chemical composition exhibit spatial and temporal variability, calibration parameterization is mandatory for different locations and seasons. One important factor contributing to the discrepancy between BA and GM measurements is the amount of material (e.g., water and/or other semi-volatile compounds) that can be evaporated from (Chang et al., 2001; Takahashi et al., 2008; Shin et al., 2011, 2012) or adsorbed on the filter or the particle-sample matrix. The filter type is among the most important contributors to the difference between the referred methods, due to their efficiency for releasing or adsorbing water or semi-volatile species such as nitrate salts, during periods of high atmospheric stability (Perrino et al., 2013). Additionally, the influence of ambient humidity conditions on different filter types can lead to non-linear hysteresis effects during the equilibration procedure of filters (Brown et al., 2006).

Price et al. (2003) found that particle bound water may be the major factor in the observed discrepancy between the gravimetric and the TEOM real time monitor, when operating at 50 °C. The water content of particles collected on filters depends on the relative humidity (RH) of the sampled air as well as the chemical composition of the particles. GM samples are collected in ambient conditions but are then weighed at ~50% RH. BA measurements on the other hand are performed in ambient conditions but in order to prevent water condensing on the filter tape, the inlet tube of BA monitors is typically heated until the moisture content of the sampled air reduced to 35% or below a specified limit, which vary and depend on the different site locations and meteorological conditions (Gobeli, 2008). These differences in sampling/measurement conditions between the BA and the GM method can significantly affect quantitative agreement between the two methods.

In the present study we report on systematic $PM_{2.5}$ measurements performed at a suburban site in Athens, Greece, during the period 2009–2012, using BA Thermo ESM Andersen FH62I-R monitors and the Gravimetric Reference method according to the EN12341 Standard. The effect of factors such as the meteorological conditions and the type of filter material used for sampling was investigated. Our analysis focuses more on $PM_{2.5}$ observations because they have not been studied so extensively as PM_{10} measurements, and in addition they exhibit increased health risks in relation to coarse particles (Kampa and Castanas, 2008). It should be noted that given the documented differences in the chemical composition of coarse and fine airborne particles in the Mediterranean area (Lazaridis et al., 2005, 2006), the comparability of BA and GM data can differ significantly for these two particle size fractions.

2. Methodology

2.1. The study area

PM measurements by GM and BA methods were performed from January 2009 to December 2012 at two co-located monitoring stations in the suburban area of Agia Paraskevi in Athens, Greece. The reference GM measurements were conducted at the DEM station (N37°59′42″.00; E23°48′57″.60) of the National Center for Scientific Research (NCSR) "Demokritos", which is part of the Global Atmosphere Watch (GAW) and ACTRIS networks. The BA measurements were performed at a National Monitoring Network (NMN) station, operated by the Hellenic Ministry of Environment, Energy & Climate Change (N37°59′42″.39; E23°49′09″.90) also situated inside the NCSR "Demokritos" campus, at a distance of 315 m from DEM station. The NCSR "Demokritos" campus is located in the north-east of the city of Athens, at the foothills of Hymettus Mountain and at an altitude of 270 m above sea level. The site is characterized by rich vegetation of pine trees. Both stations have a distance of approximately 400 m from the Hymettus Peripheral road and 1 km from a regular traffic avenue (Fig. 1). The background nature of this suburban area ensures that parallel measurements were conducted under the same meteorological conditions and were influenced from the same sources. The uncertainty introduced to the data by the variability in concentrations due to the small distance between the two sites can be therefore considered insignificant. The area exhibits low PM concentrations (Eleftheriadis et al., 2014; Triantafyllou and Biskos, 2012; MinEnv,

2013; Kostenidou et al., 2015) and is representative of suburban background conditions.

2.2. Measurements and methods

Two BA instruments (Thermo ESM Andersen FH62I-R) were used for the monitoring of PM_{10} and $PM_{2.5}$ concentrations. The instruments utilize the radiometric principle of beta attenuation by a 2-beam compensation method. Particles are collected at a flow rate of 1 m^3 h^{-1} on a glass-fiber filter tape and exposed to a low level (50 mCi) beta radiation emitted by an ^{85}Kr source. The radiation attenuated by the particle loadings on the filter tape is continuously monitored and converted to PM mass concentrations (Baron and Willeke, 2001). The accumulated particle mass is also monitored. The concentration range of the BA monitors is set in the range 0–1 $\times$ 10^3 μg m^{-3}. Measurements at the Ministry of Environment stations employed for this study, were obtained by BA monitors equipped with a heated inlet maintained at 28 ± 2 °C.

PM_{10} and $PM_{2.5}$ GM measurements were performed using custom-made low-volume reference gravimetric samplers. Three types of filters were used throughout the measurement period: teflon, quartz and glass fiber. Filters were weighed before and after sampling by a microbalance (Sartorius Model BP 211 D) of 0.01 mg accuracy. Prior to weighing, the filters were conditioned at 20 ± 1 °C and 50 ± 5% RH for 48 h, according to the EN12341 (1998) guidelines. PM concentrations generally corresponded to 24-h sampling, but in some cases we used longer sampling periods of up to 72 h.

Meteorological conditions (i.e., temperature, RH, as well as wind speed and direction) were monitored throughout the measurement period. Back-trajectory analysis by HYSPLIT model (Draxler and Rolph, 2013) was also used in order to investigate the effect of the origin of air masses arriving at the stations. The near real-time concentrations measured by the BA monitors were averaged based on the duration of the GM measurements, and classified in two main groups depending on the sampling period. The first group included measurements conducted during the cold period (i.e., from 16 October to 15 April) and the second during the warm period (i.e., from 16 April to15 October).

3. Results and discussion

3.1. $PM_{2.5}$ measurements

3.1.1. Concentration levels

$PM_{2.5}$ concentrations measured by the GM and BA methods (referred to as the GM $PM_{2.5}$ and BA $PM_{2.5}$ measurements, respectively, from this point onwards) are presented in Fig. 2. In general, the two data sets exhibited similar temporal variation. The measured $PM_{2.5}$ concentration levels ranged from 2 to 82 μgm^{-3} for the GM method, and from 5 to 52 μgm^{-3} for the BA method. Two of the three highest concentrations (>50 μgm^{-3}) measured by the GM method corresponded to Sahara dust long range transport episodes. BA measurements were not available for these days.

The mean PM concentrations measured by the GM and BA methods for each year are shown in Table 1. The BA $PM_{2.5}$/GM $PM_{2.5}$ concentration ratio for the entire study period indicated that the BA levels were on average approximately 30% higher than the respective GM values (ANOVA, p-value < 0.05). This overestimation is more pronounced during the warm period. The BA $PM_{2.5}$ to GM $PM_{2.5}$ relationship exhibited significant day-to-day variability, suggesting that various factors may affect the measured concentration levels.

3.1.2. Correlation between BA and GM measurements

Good correlation was observed between BA $PM_{2.5}$ and GM $PM_{2.5}$, as depicted in Fig. 3a (Pearson coefficient was 0.79 for the entire period, and 0.72 and 0.84 for the warm and cold seasons, respectively). Best linear fits were calculated for all data and each period separately: BA = 0.71 G M + 6.2 (entire period); BA = 0.62 G M + 8.3 (warm period); BA = 0.77 G M + 4.7 (cold period). Although the trends in the warm- and cold-period data are similar, the differences in the fitted equations highlight the need for



Fig. 1. Map showing the location of the monitoring stations operated by NCSR "Demokritos" (DEM Station) and the National Monitoring Network (NMN Station) at the suburban area of Agia Paraskevi in Athens, Greece.

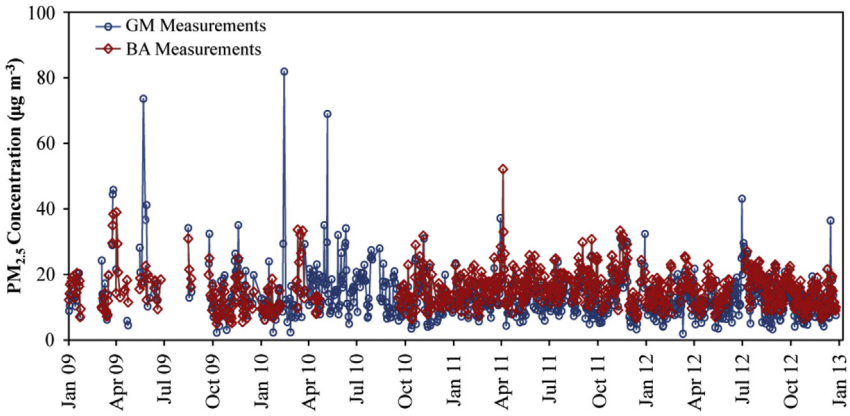


Fig. 2. Time series of $PM_{2.5}$ mass concentrations measured by gravimetric (blue circles) and beta attenuation (red diamonds) methods at Agia Paraskevi from 2009 to 2012. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Descriptive statistics for the GM $PM_{2.5}$ and BA $PM_{2.5}$ concentrations measured at Agia Paraskevi from 2009 to 2012.

		GM $PM_{2.5}$		BA $PM_{2.5}$		BA $PM_{2.5}$/GM $PM_{2.5}$
	Year	Number of samples (N)	Mean ± St. Deviation (μg m^{-3})	Number of samples (N)	Mean ± St. Deviation (μg m^{-3})	Mean ± St. Deviation
Entire Period	2009	95	16.5 ± 10.5	102	14.7 ± 6.7	1.02 ± 0.47
	2010	212	14.6 ± 9.2	103	14.1 ± 6.0	1.25 ± 0.46
	2011	254	13.0 ± 5.3	255	17.0 ± 5.6	1.36 ± 0.32
	2012	324	12.4 ± 5.5	285	14.7 ± 4.3	1.33 ± 0.37
	2009–2012	885	13.5 ± 7.3	745	15.4 ± 5.5	1.29 ± 0.40
Warm Period	2009	29	18.6 ± 14.4	31	15.8 ± 5.5	1.19 ± 0.63
	2010	116	16.0 ± 8.9	23	12.9 ± 3.3	1.15 ± 0.44
	2011	132	12.4 ± 4.4	132	17.1 ± 4.3	1.45 ± 0.37
	2012	163	13.2 ± 6.0	136	15.9 ± 4.4	1.42 ± 0.47
	2009–2012	412	14.0 ± 7.5	292	16.2 ± 4.5	1.40 ± 0.46
Cold Period	2009	66	15.5 ± 8.1	71	14.3 ± 7.2	0.95 ± 0.36
	2010	96	12.9 ± 9.4	80	14.5 ± 6.6	1.28 ± 0.47
	2011	122	13.8 ± 6.1	123	17.0 ± 6.8	1.27 ± 0.23
	2012	161	11.5 ± 4.8	149	13.6 ± 4.0	1.24 ± 0.23
	2009–2012	445	13.0 ± 7.0	423	14.9 ± 6.1	1.21 ± 0.33

seasonal calibration of BA data through reference gravimetric measurements.

Correlation between BA $PM_{2.5}$ and GM $PM_{2.5}$ data was also examined depending on the filter type used in the GM measurements (Fig. 3(b)–(d)). A slightly better correlation was observed when glass fiber filters were used for the GM samples (estimated Pearson correlation coefficients were 0.82 and 0.80 for the Teflon and the quartz filters, respectively). This agreement was anticipated, considering that the BA monitor employs a glass fiber filter tape. The equation that fitted the data best in the case of glass fiber filters presented the highest slope (BA = 0.79 G M + 5.3). The corresponding best linear fits in the case of Teflon and quartz filters were BA = 0.75 G M + 6.1 and BA = 0.64 G M + 4.5, respectively.

For all filter types, better correlations were obtained for the cold rather than the warm period. The largest difference between cold and warm period correlation relationships was observed for glass fiber samples, followed by quartz samples. These discrepancies may be attributed to the different water uptake of each filter type. Glass fiber filters, which are used as standard filter material in BA measurements, are moisture sensitive.

Quartz filters on the other hand present lower water uptake, whereas Teflon filters are generally hydrophobic (Baron and Willeke, 2001). The results indicate that the selection of the filter substrate may influence the measured PM levels and that filter material should be taken into account when calibrating BA with GM measurements.

The relation between BA $PM_{2.5}$ and GM $PM_{2.5}$ measurements was also examined in relation to the origin of air masses arriving at the site, which is an indicator of different types of aerosols (with respect to chemical composition and size distribution). Sahara dust long-range transport events were identified during only 4% of the measurement days and did not affect the relationships between BA PM and GM PM measurements. The rest of the measurements were separated into three groups depending on the conditions: (i) days of calm conditions, representative of local aerosol (with more than 50% of hourly wind speed values below 1 ms^{-1}), (ii) days with south/west wind directions representative of polluted aerosols coming from the city center, and (iii) days with north/east wind directions representative of "cleaner" aerosols coming from the sea. No change in the BA/GM relationships was observed, suggesting that the linear regression equations obtained for warm and cold period are representative for the investigated site, and may be used for calibration of BA monitors.

3.1.3. *Effect of meteorological and sampling conditions*

To further investigate whether the seasonal variability in the differences between BA and GM measurements is attributed to water evaporation, we tried to estimate the water content of the

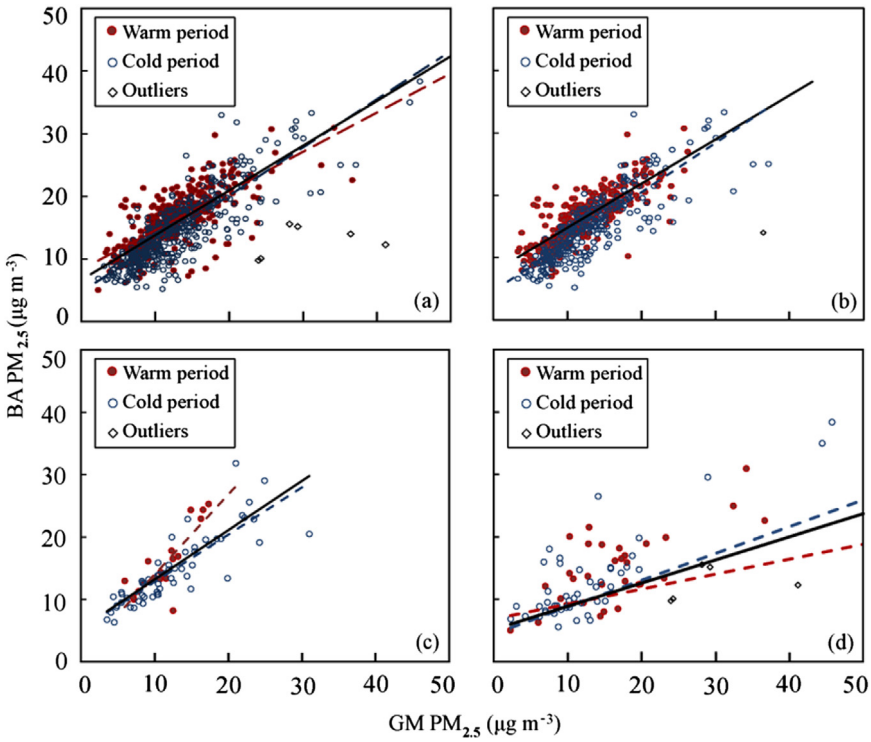


Fig. 3. Linear correlation between BA $PM_{2.5}$ and GM $PM_{2.5}$ measurements for (a) all the data, and when (b) Teflon, (c) Glass fiber and (d) Quartz were used in the GM measurements. Red and blue circles correspond to measurements during the warm and the cold period, respectively. Outliers are depicted by diamond-shaped data points. The black line represents the linear fit for all data, while red and blue dashed lines are the fitted equations to the warm and cold season data, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

samples. This depends on the ability of the particles to hold water under given RH conditions, i.e. their hygroscopicity (Colbeck, 2008; Bezantakos et al., 2013), which mainly depends on their chemical composition: inorganic particles are more hygroscopic whereas soot and organic particles can take up significantly lower amounts of water. In addition, the material of the sampling medium may also take up water during sampling. Glass fiber and quartz filters have a slightly alkaline surface and are therefore affected by humidity, while membrane filters such as Teflon exhibit lower moisture absorption. The amount of liquid water in the atmosphere that can be absorbed by the particles and the filters is:

$$e = RH \cdot e_s \quad (1)$$

where RH is the mean relative humidity (%) in the atmosphere during sampling, and e_s the maximum water vapor pressure (mbar). For a given temperature, T (K), e_s can be estimated by the modified empirical Magnus–Tetens relation (Salby, 1996):

$$\log e_s = 9.4041 - \frac{2354}{T} \quad (2)$$

The estimated water vapor pressure can be normalized by the GM $PM_{2.5}$ measured concentration, and the resulting value represents the potentially available water for interaction with each unit of sampled mass.

The monthly averaged BA $PM_{2.5}$ to GM $PM_{2.5}$ concentration ratio, along with the respective temperature and relative humidity conditions, is shown in Fig. 4. The BA $PM_{2.5}$/GM $PM_{2.5}$ ratio was consistently higher than 1.3 when the RH decreased below 60% and the temperature was higher than 20 °C. The lowest values of the BA $PM_{2.5}$/GM $PM_{2.5}$ concentration ratio (i.e., around 1.2) were observed during the cold period when ambient temperature was around or below 10 °C and RH was above 70%. Given that a heating inlet was employed at the BA monitor (in order to bring the sample to a standard temperature of 28 °C) and not in the GM measurements, the lower values of the ratio during the cold period may be related to the evaporation of water and loss of volatile aerosol compounds during heating of the sampled air. During the warm period, volatile aerosol compounds are not present when sample air enters the inlet due to high ambient air temperatures. As shown in Fig. 4, the normalized water vapor pressure exhibits a seasonal variation that is very similar with that of the BA $PM_{2.5}$/GM $PM_{2.5}$ ratio. This similarity indicates that the samples' water content strongly affects the discrepancies observed between BA and GM measurements.

The relationship between the BA $PM_{2.5}$/GM $PM_{2.5}$ ratio and the normalized water vapor pressure, for the entire data set and for the measurements using the different filter types, is shown in Fig. 5. BA to GM concentration ratios exhibited an increasing trend with normalized water vapor pressure in all cases. It is interesting to note that the curve fitted through the measurements is steeper when Teflon filters (which are highly hydrophobic) are used in the GM measurements (Fig. 5b).

Teflon is the least moisture sensitive filter material and it is thus less affected by increased water content, in comparison to the glass fiber filter tape used in BA monitor. The return of the filters to the reference conditions at the weighing room appears to remove the excess moisture, while the heating system or the temperature difference between BA monitor and ambient atmosphere is not as efficient. The BA $PM_{2.5}$/GM $PM_{2.5}$ concentration ratio is therefore found to be above unity. The filter material is partly affecting the value of this ratio. As indicated by the data shown in Fig. 5c, when GM measurements are conducted with glass fiber filters, which is the same material used in the BA monitors, the effect of the differential water absorption on the aerosol sample is reduced.

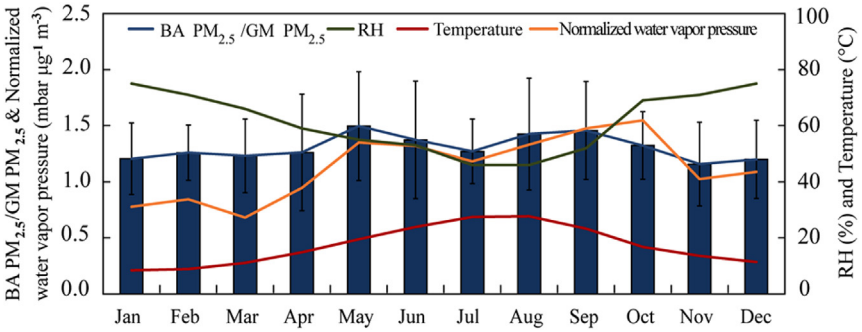


Fig. 4. Variation of the monthly averaged BA $PM_{2.5}$ to GM $PM_{2.5}$ concentration ratio (bars and blue line) and of the corresponding temperature (red line), relative humidity (green line) and the water vapor pressure normalized by $PM_{2.5}$ concentrations (orange line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

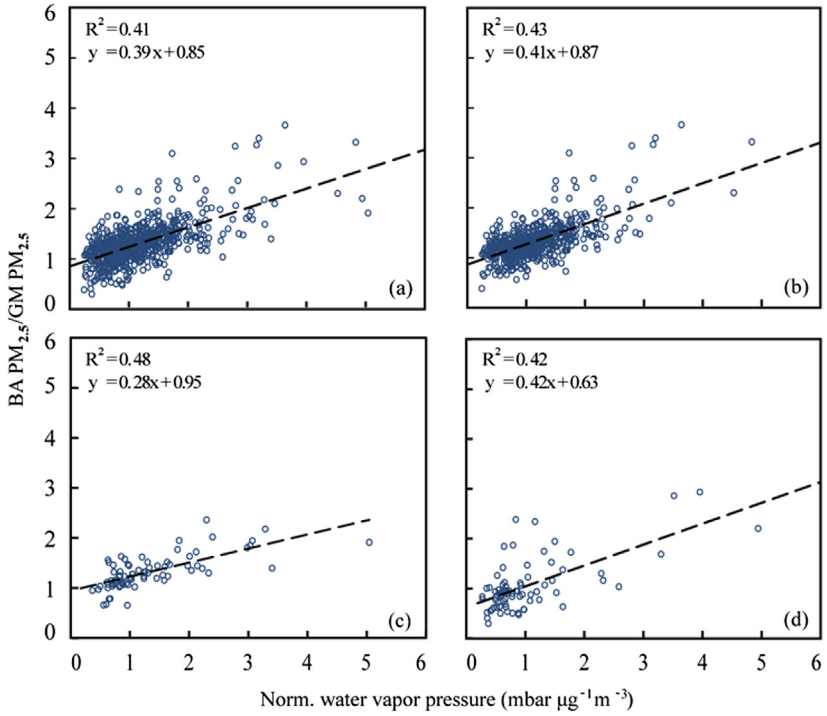


Fig. 5. Correlation between the BA $PM_{2.5}$/GM $PM_{2.5}$ ratio and the normalized water vapor pressure, for (a) the entire data set and separately when (b) Teflon, (c) glass fiber, and (d) quartz filters were used in the GM measurements. The dashed line show best fit regression models.

In order to evaluate the other issues of possible gain or loss of water and/or other species, further analysis of the parameters involved was conducted. As already shown, temperature or RH alone cannot explain the discrepancy between BA and GM measurements, as well as the normalized available water content does (Fig. 4). Temperature is a critical parameter for semi volatile species like ammonium nitrate as expected from basic thermodynamic considerations (Dassios and Pandis, 1999). Pandolfi et al. (2012) observed that temperature is the only parameter showing a relatively high correlation with NO_3. The temperature range between 15 and 22 °C has been also shown to play a role in the ambient levels of the semi volatile species according to model calculations and ambient data for Riverside CA, US (Aw and Kleeman, 2003). Although these effects are difficult to quantify, conditioning of filters and PM samples at 20 °C and 50% RH sets some reference conditions at which the discrepancy between BA and GM measurements can be evaluated, thereby providing a systematic way of its characterization.

Useful information about the structure of the variability of the BA to GM ratio can be derived when we examine this ratio against RH over three temperature ranges 0–10 °C, 11–22 °C and greater than 22 °C. Scatter plots of this exercise are shown in Fig. 6.

Despite the high scatter we observe that the linear fit lines display no significant trend of the discrepancy at high and low temperatures and that the mean value across the whole of the RH range is similar to the ratios shown in Table 1, with BA measurements constantly overestimating the GM $PM_{2.5}$ concentration. When the mean temperature is within the range 11–22 °C we have a significant number of data where BA measurements underestimate the $PM_{2.5}$ concentration at high RH and a linear trend

becomes significant ($p \ll 0.01$). This trend appearing for the temperature range studied can be attributed to the loss of nitrate. This assumption is supported by the fact that ammonium nitrate salts are found in the urban atmosphere in significant concentrations. Other semi volatile species can also cause an equivalent discrepancy under these conditions (see discussion below).

The dataset for the Teflon measurements shown in Fig. 7 is evaluated in the same way against the normalized water content over the same temperature ranges. The temperature range between 11 and 22 °C does not display a distinct behaviour as that in the measurements shown in Fig. 6. The available water does not seem to depend on RH for all three temperature ranges, while the overestimation of $PM_{2.5}$ concentration by the BA measurements is always correlated with liquid water, with the correlation coefficient becoming stronger for higher temperatures. This feature indicates that BA systems retain or accumulate liquid water due to condensation on the measured sample, while conditioning of filters during GM measurements removes liquid water more efficiently. On the other hand, the difference in the water content does not explain alone the dependence of the BA/GM ratio to RH as shown in Fig. 6b. It is reasonable to explore the possibility of evaporation/condensation of other semi volatile species such as organics and ammonium nitrate, which may be in transition from particulate to gaseous phase at these temperature and RH ranges. Especially for ammonium nitrate, the particle phase is favored at high RH and lower temperatures. During our study period, estimated ammonium nitrate levels inferred from earlier measurements at the same site (Eleftheriadis et al., 2014 reporting total NH_4 and NO_3 to be on average 1.3 μg m^{-3} and 1.4 μg m^{-3}, respectively) are expected in general to be below 2 μg m^{-3}. This in turn is a 10–15% of the $PM_{2.5}$ mass concentration. During periods when maximum ammonium nitrate concentrations are expected in the particulate phase due to high RH (Dassios and Pandis, 1999), dissociation due to BA inlet heating and retention in the GM filter at ambient and conditioning temperatures < 20 °C can explain the lower BA/GM ratios observed in Fig. 6b. The temperature difference between the heated inlet

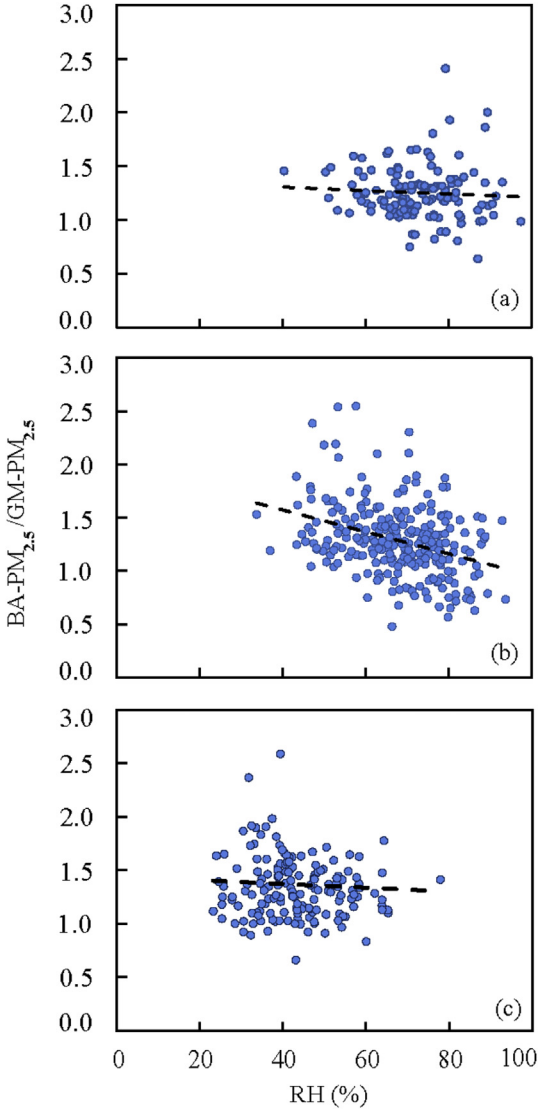


Fig. 6. Correlation between the BA $PM_{2.5}$/GM $PM_{2.5}$ ratio and the relative humidity when Teflon filters were used in the GM measurements and the ambient temperature was (a) less than 10 °C, (b) between 11 and 22 °C, and (c) higher than 22 °C. The dashed line show best-fit regression models.

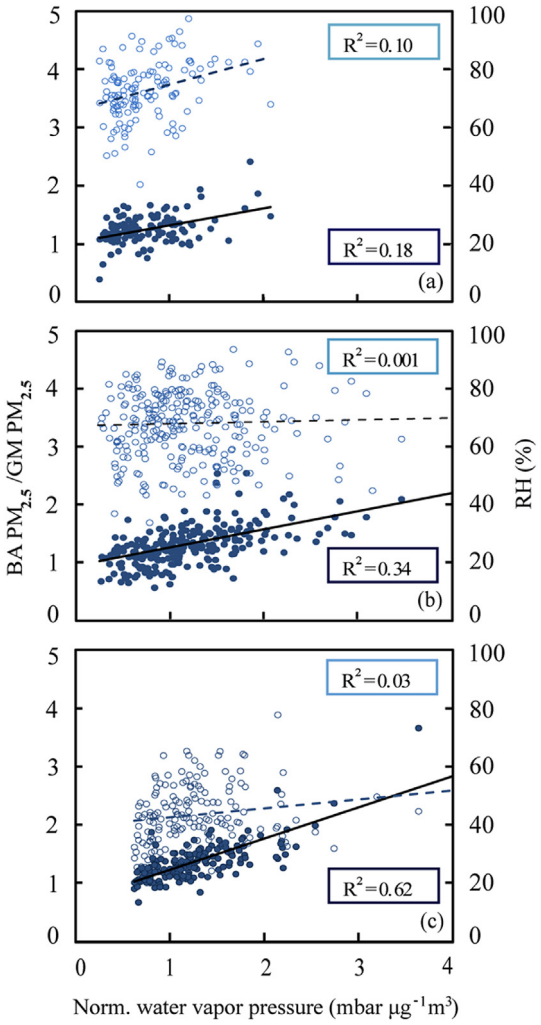


Fig. 7. Correlation between the normalized water vapor pressure and the BA $PM_{2.5}$/GM $PM_{2.5}$ ratio (blue solid circles), or the relative humidity (blue open circles), when the ambient temperature was (a) less than 10 °C, (b) between 11 and 22 °C, and (c) higher than 22 °C. The black solid line and the blue dashed line represent best linear fit for the correlations between the normalized water vapor pressure and the BA $PM_{2.5}$/GM $PM_{2.5}$ ratio and the relative humidity, respectively. These correlations include the $PM_{2.5}$ samples collected on Teflon filters. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(28 °C) and the equilibrium temperature for filters during gravimetric analysis can induce an increased evaporation of NH_4NO_3 as well as aerosol particle water during BA measurements while GM measurements remain unaffected.

The analysis presented here for PTFE filters could not be repeated for an equivalent number of data for the GF and Quartz filter due to a smaller number of samples on these filters. The data available display either similar behaviour with PTFE filters or absence of correlation.

Based on the above observations, it is proposed that when equivalence tests are conducted the results are checked against the normalized water content in order to remove seasonal discrepancies external to the conditioning of the aerosol samples. A method of drying the incoming aerosol before a BA system instead of heating would also facilitate towards reducing the effect of water vapor condensation and loss of aerosol due to heating.

3.2. PM_{10} measurements

A less extensive data set with BA and GM PM_{10} concentration measurements was also obtained during the studied period. All GM PM_{10} measurements were conducted with Teflon filters. Descriptive statistics of this data set are provided in Table 2. Mean PM_{10} concentrations measured by the two methods exhibited an average difference of 10% for the entire period (ANOVA, p-value = 0.05). Very good correlation was observed between BA PM_{10} and GM PM_{10} mass concentration measurements (Fig. 8). Different linear regression equations were calculated for the warm and cold season measurements, highlighting the seasonality of BA to GM relationship.

The obtained results are comparable to those reported by Lazaridis et al. (2008) who have also observed a slight over prediction of the PM_{10} concentrations measured by a BA monitor during measurements on the island of Crete, Greece (mean BA/GM ratio around 1.2). While BA and GM methods for PM_{10} measurements are in good agreement, the application of calibration equations, derived through parallel gravimetric measurements at both cold and warm season, can certainly enhance the reliability of the measurements obtained by BA monitors.

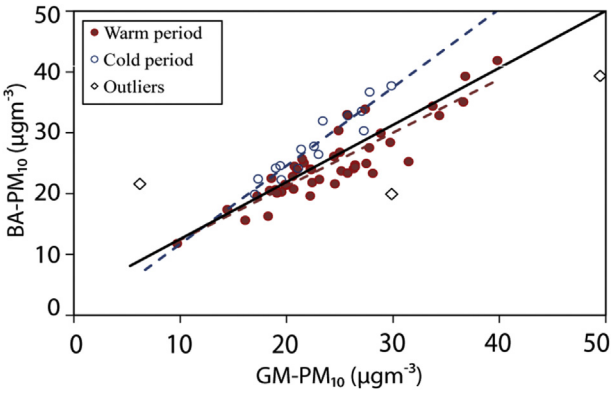


Fig. 8. Correlation between BA PM_{10} and GM PM_{10} measurements. Red and blue circles correspond to measurements during the warm and the cold period, respectively. Outliers are depicted by the black diamonds. The solid black line (BA = 0.77 G M + 4.1, R^2 = 0.72) represents the best linear fit for the entire data set. The red dashed line (BA = 0.88 G M + 3.6, R^2 = 0.80) and the blue dashed line (BA = 1.50 G M − 6.7, R^2 = 0.88) represent best linear fits for the warm and cold period, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4. Conclusions

This work presents a comparison of $PM_{2.5}$ and PM_{10} concentration measurements performed by a single type of beta attenuation monitors (Thermo ESM Andersen FH62I-R monitors) and reference gravimetric samplers at a suburban site in Athens, Greece, during a period of four years. Given that the majority of previous similar studies used PM_{10} measurements, here we focus on $PM_{2.5}$ measurements that have greater human health relevance. BA and GM measurements showed in general similar temporal variation, for both PM_{10} and $PM_{2.5}$ fractions, with the former being systematically higher. This difference was more pronounced for $PM_{2.5}$ (on average 30%) compared to the PM_{10} (on average 10%) measurements. The correlation between BA and GM measurements was higher during the cold period for both size fractions, with Pearson coefficient being 0.84 for $PM_{2.5}$ and 0.95 for PM_{10} in the cold period and 0.72 for $PM_{2.5}$ and 0.90 for PM_{10} in the warm period.

The seasonality in the differences between BA and GM measurements could be largely explained by a strong effect of temperature and humidity. The BA $PM_{2.5}$/GM $PM_{2.5}$ ratio increased with atmospheric water vapor pressure, indicating that the water content of the collected samples has a strong effect on the observed discrepancy between BA and GM measurements. The filter material used in the GM measurements was identified as another important factor for this discrepancy. The strongest correlation between BA $PM_{2.5}$ and GM $PM_{2.5}$ measurements and the one having a fitted line with a slope closest to unity were obtained when GM samples were collected on filters made of glass fiber, which is the same material used in the filter tapes in the BA measurements. The highly hydrophobic Teflon filters were less affected by moisture in comparison to quartz and glass fiber filters, yielding larger discrepancies between BA and GM measurements when the atmospheric water vapor pressure increased.

Our results show that differences between BA and GM measurements can vary significantly when determining PM_{10} and $PM_{2.5}$ concentrations, and that relationship between the two observation methods ought to be determined for each site and season. The measurements also highlight the importance of reporting the meteorological conditions and operational characteristics (i.e., filter materials and sampling temperatures of the BA monitor inlet and unit) along with the measurements, since they can strongly affect the results. Considering that inter-comparisons are scarce for the $PM_{2.5}$ fraction, additional measurements, in different climatic regions, are required to obtain a better overall picture for the effects of liquid water condensation and loss of semi volatile species.

Table 2
Descriptive statistics of PM_{10} concentrations measured at Agia Paraskevi by beta-attenuation (BA) and gravimetric method (GM) for the years 2011 and 2012.

Year	GM PM_{10}		BA PM_{10}		BA PM_{10}/GM PM_{10}
	Number of samples (N)	Mean ± St. Deviation (μg m^{-3})	Number of samples (N)	Mean ± St. Deviation (μg m^{-3})	Mean ± St. Deviation
2011–2012	95	23.4 ± 14.9	67	25.1 ± 6.5	1.10 ± 0.33
2011 (warm period)	48	24.4 ± 7.5	48	24.9 ± 6.3	1.08 ± 0.38
2012 (cold period)	47	22.4 ± 19.8	19	25.7 ± 7.1	1.18 ± 0.15

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